



Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

**Lab 01: GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation**

DevOps

Name : Eman Khalid

Reg no : 23-BCS-026

Course Code: CS-649

Section: A

GitHub username: EmanKhalid27

Submitted To: Sir Waqas Saleem

Department: Computer Science

Lab Objective

In this lab, you will create and correctly configure your GitHub account, register yourself in the Student Marks Portal, install Ubuntu Server in VMware Workstation Pro, verify your required Linux identity, and connect to the server remotely from Windows.

Task List

Getting Started

- Task 1: Create and Configure Your GitHub Account
- Task 2: Register in the Student Marks Portal
- Task 3: Download VMware Workstation Pro and Ubuntu Server
- Task 4: Create the Ubuntu Server Virtual Machine
- Task 5: Choose the Installation Language
- Task 6: Configure the Keyboard Layout
- Task 7: Choose the Ubuntu Server Installation Type
- Task 8: Configure the Network
- Task 9: Configure Storage
- Task 10: Select the Installation Disk
- Task 11: Review the Partitions
- Task 12: Confirm the Storage Changes
- Task 13: Configure the Ubuntu Profile
- Task 14: Complete the Software Installation
- Task 15: Reboot and Verify the Ubuntu Identity
- Task 16: Find the Ubuntu Server IP Address
- Task 17: Connect to Ubuntu Server from Windows
- Task 18: Prepare Lab1 Solution PDF

Getting Started

- Screenshot showing the created directory structure as "lab01_folder_structure.png".

 > This PC > Documents > DevOps > Labs > Lab01 > screenshots

Read the complete task list before beginning the installation.

Task 1: Create and Configure Your GitHub Account

- Screenshot clearly showing your profile name and GitHub username as "github_profile.png".



- Screenshot showing the completed Name field as "github_actual_name.png".

Public profile

Name

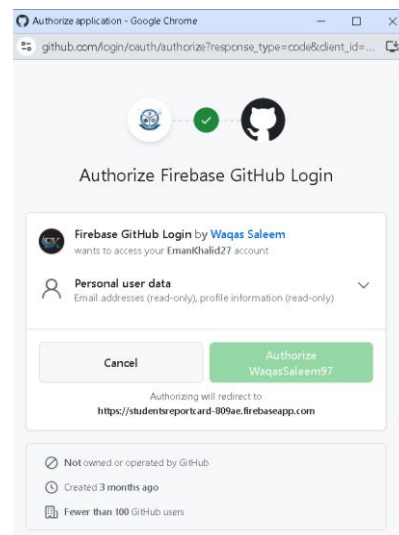
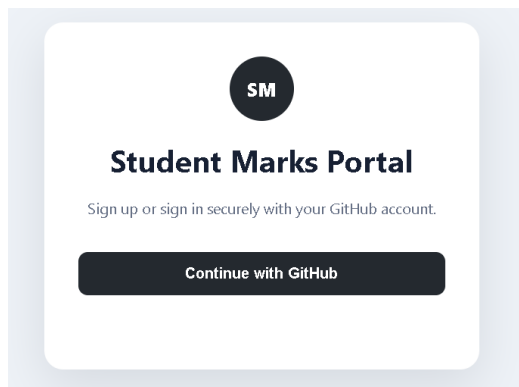
Eman Khalid

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

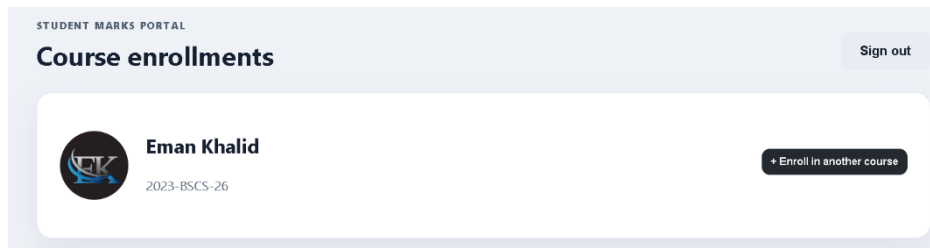
Task 2: Register in the Student Marks Portal

Open the Student Marks Portal.

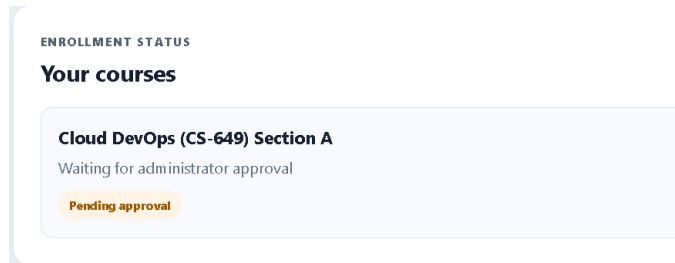
- Screenshot of the portal login page as "portal_login_page.png".



- Screenshot showing your GitHub profile information inside the portal as "portal_github_profile.png".

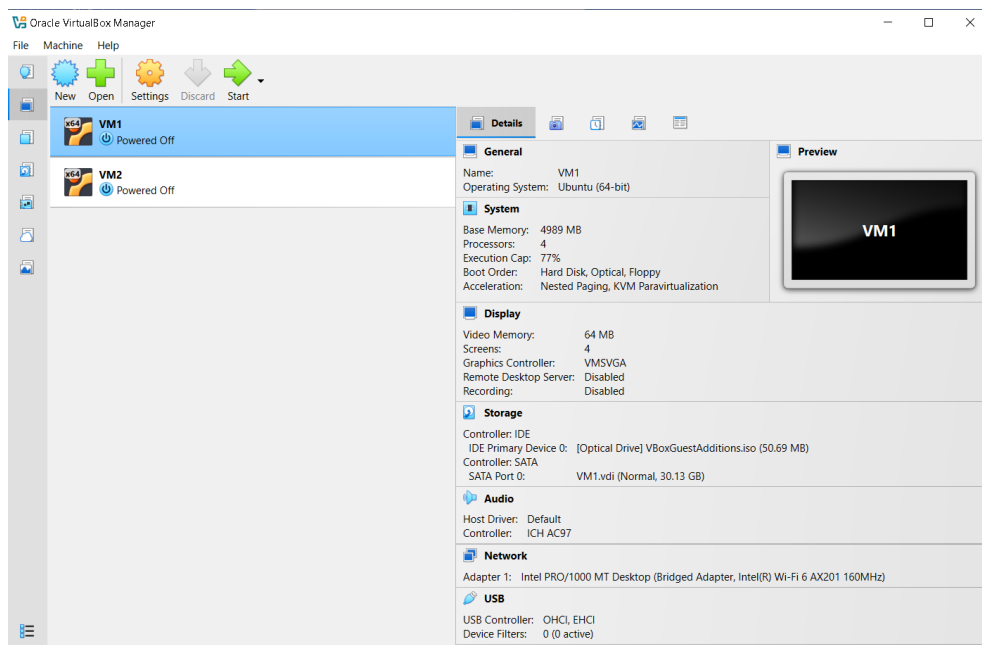


- Screenshot of the submitted or pending enrollment as "portal_enrollment_submitted.png".



Task 3: Download VMware Workstation Pro and Ubuntu Server

- Screenshot showing VMware Workstation Pro running as "vmware_workstation.png".



- Screenshot showing the downloaded Ubuntu Server ISO filename as "ubuntu_server_iso.png".

ubuntu-26.04.1-live-server-amd64.iso 10/1/2026 6:06 PM Disc Image File 2,859,240 ...

Confirm that at least 20 GB of free storage is available for the Ubuntu Server virtual machine.

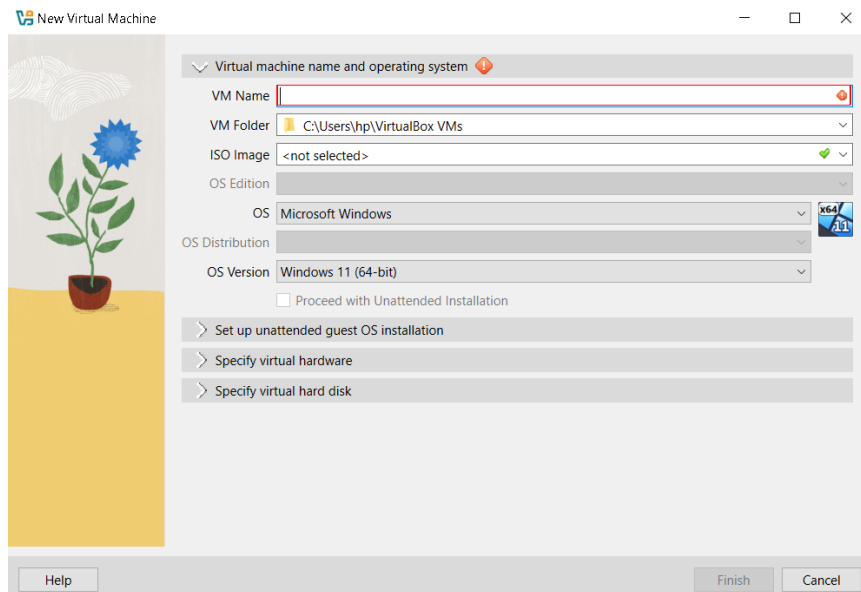
- Screenshot showing the available storage as "available_storage.png".



Task 4: Create the Ubuntu Server Virtual Machine

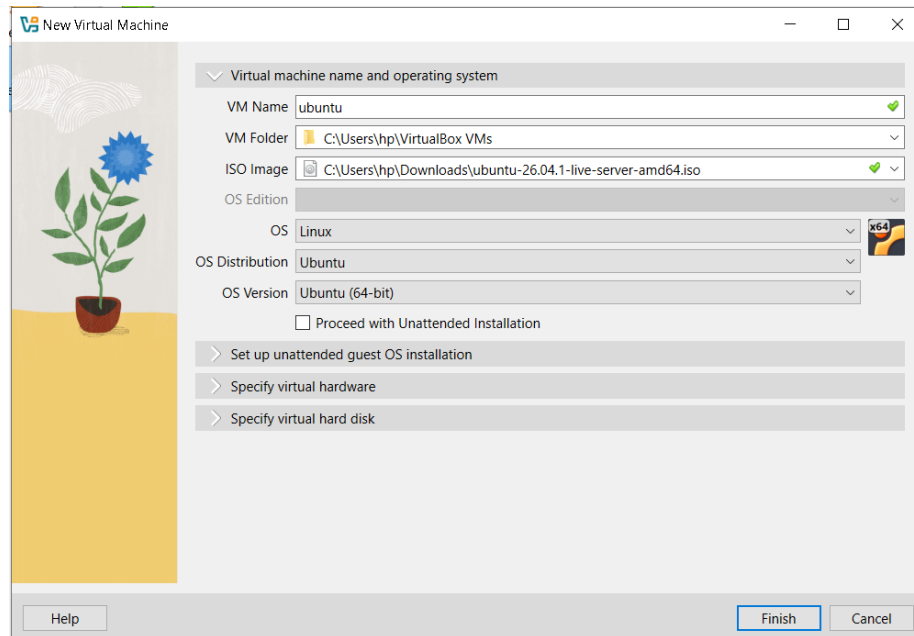
Open VMware Workstation Pro and select Create a New Virtual Machine.

- Screenshot of the New Virtual Machine Wizard as "vm_creation_wizard.png".



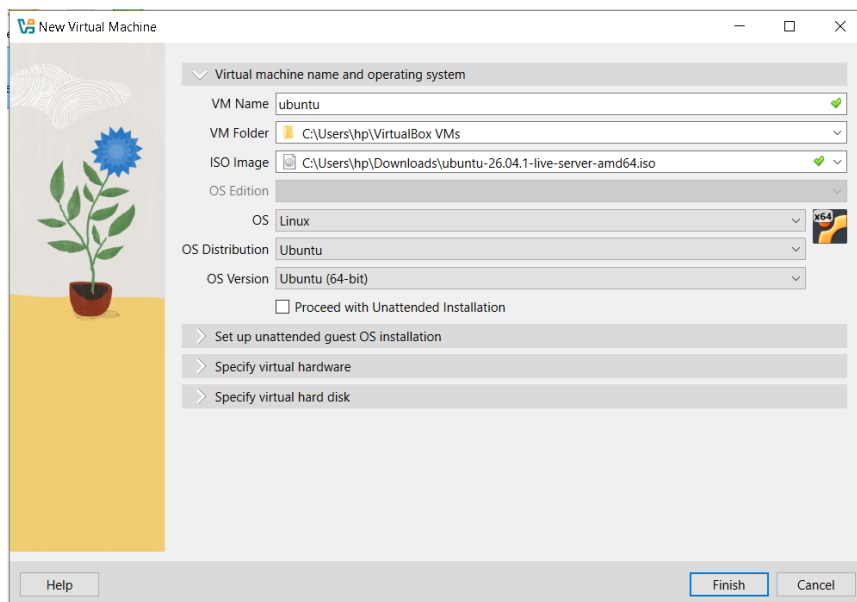
Select Typical (recommended) as the virtual machine configuration.

- Screenshot showing this selection as "vm_typical_configuration.png".

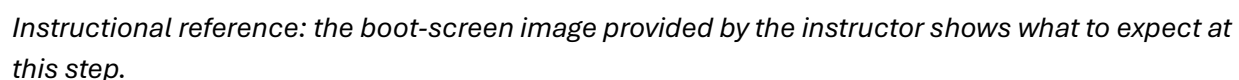
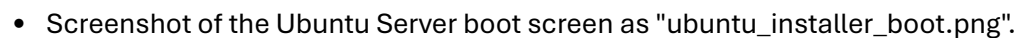


Select Installer disc image file (ISO) and browse to the Ubuntu Server ISO downloaded in Task 3.

- Screenshot showing the selected Ubuntu ISO path as "vm_ubuntu_iso_selected.png".

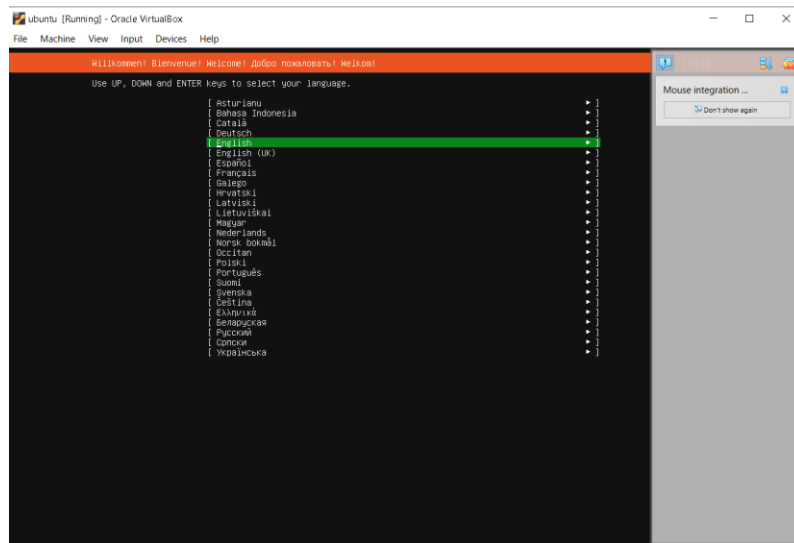


- Power on the virtual machine and wait for the Ubuntu Server installer to start.

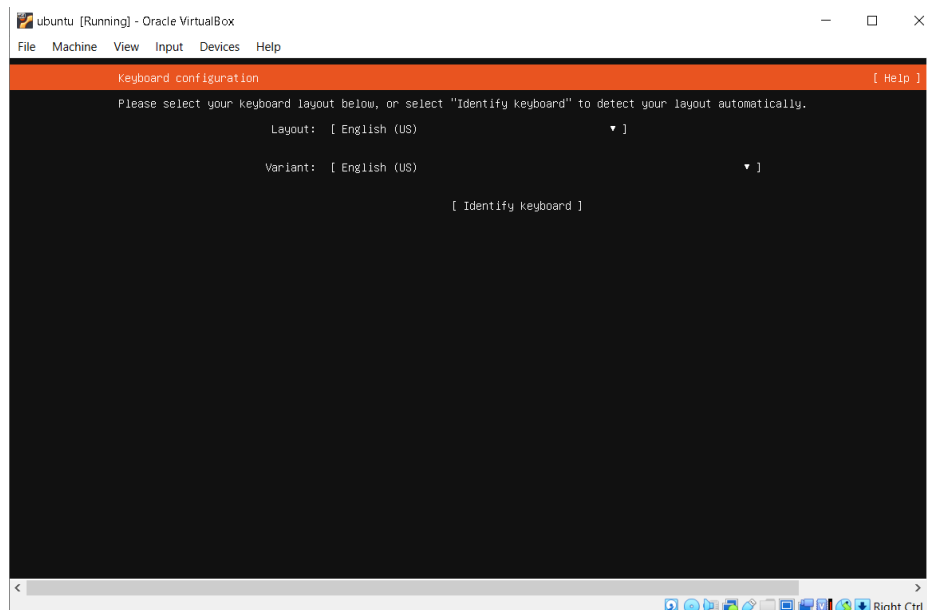


Task 5: Choose the Installation Language

- Screenshot showing the selected language as "ubuntu_language.png".



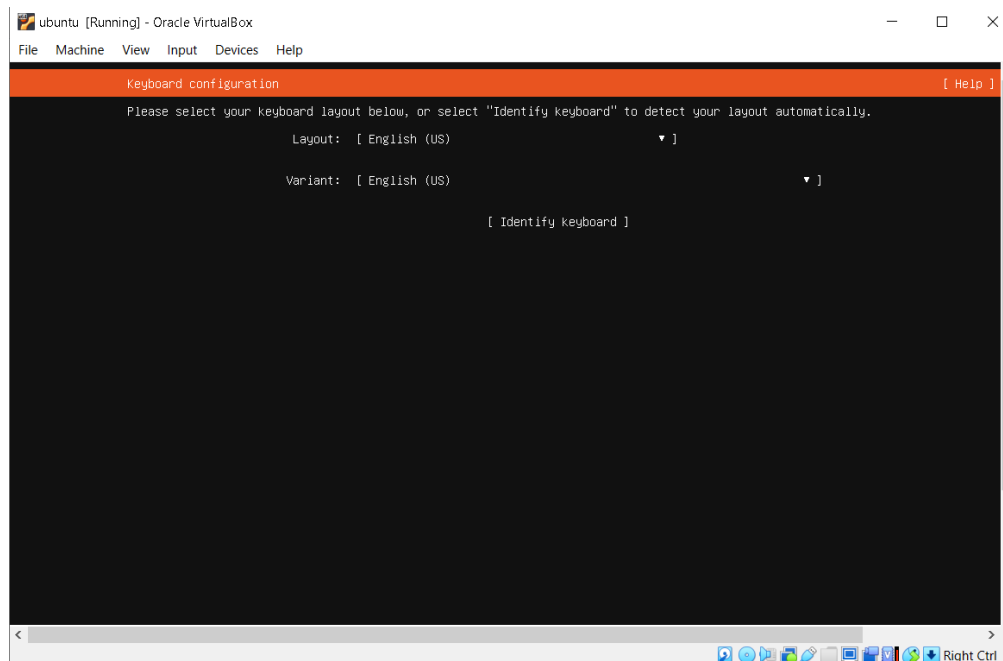
Task 6: Configure the Keyboard Layout



- Screenshot showing the selected keyboard layout as "ubuntu_keyboard_layout.png".

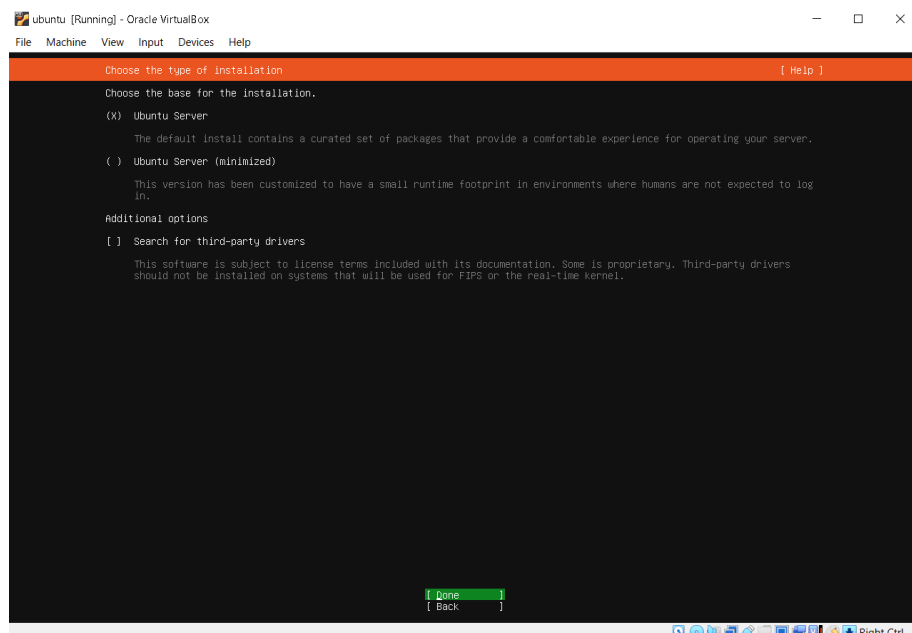
Select the appropriate keyboard variant.

- Screenshot showing the selected keyboard variant as "ubuntu_keyboard_variant.png".



Select Done to continue.

Task 7: Choose the Ubuntu Server Installation Type

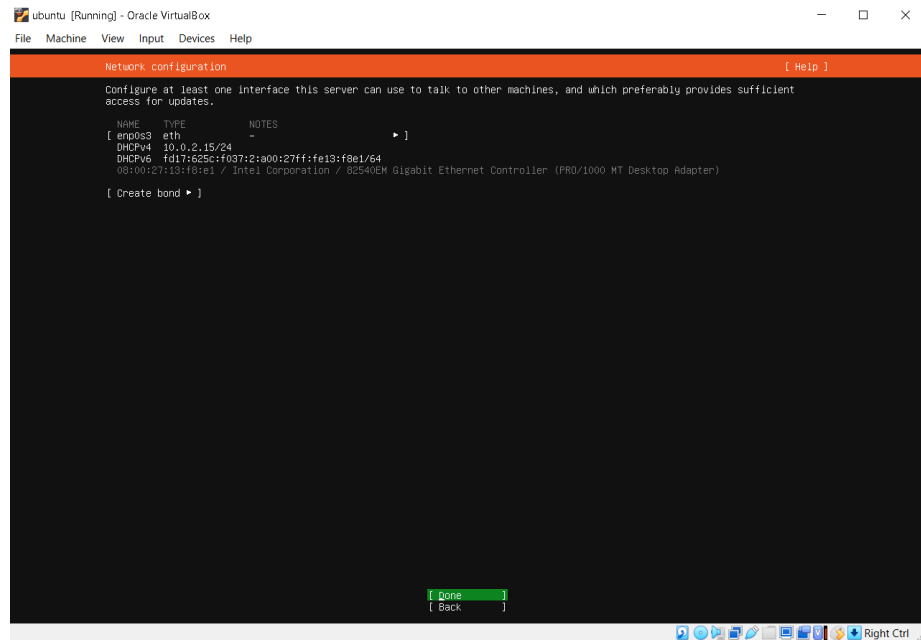


- Screenshot showing the selected installation type as "ubuntu_installation_type.png".

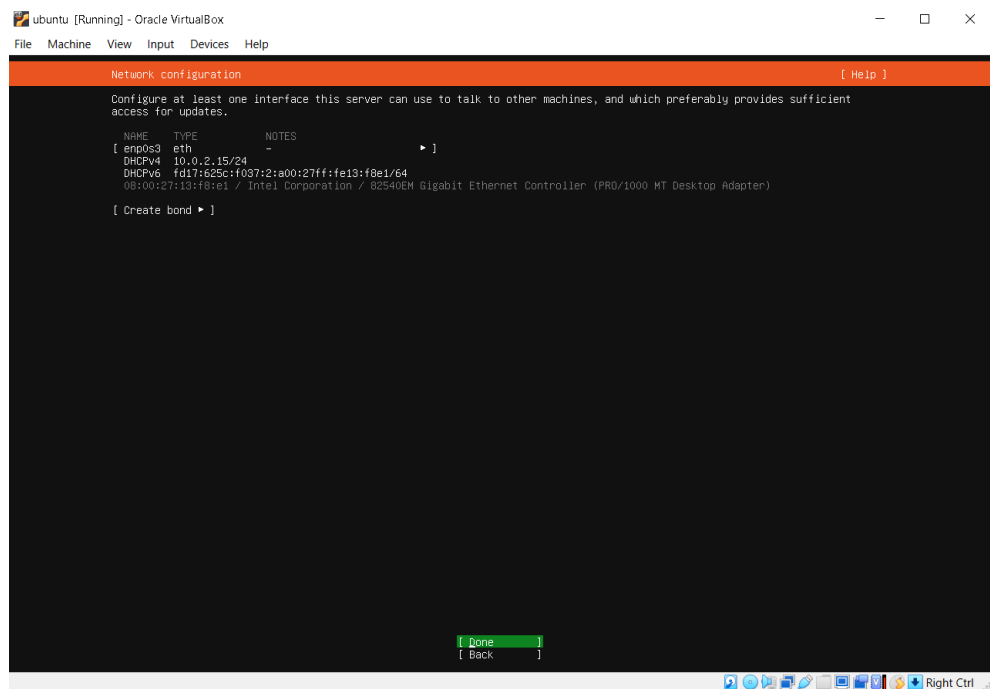
Continue to the next installer screen.

Task 8: Configure the Network

- Screenshot showing the detected network interface as "ubuntu_network_interface.png".



Confirm that the interface has received an IP address automatically through DHCP.

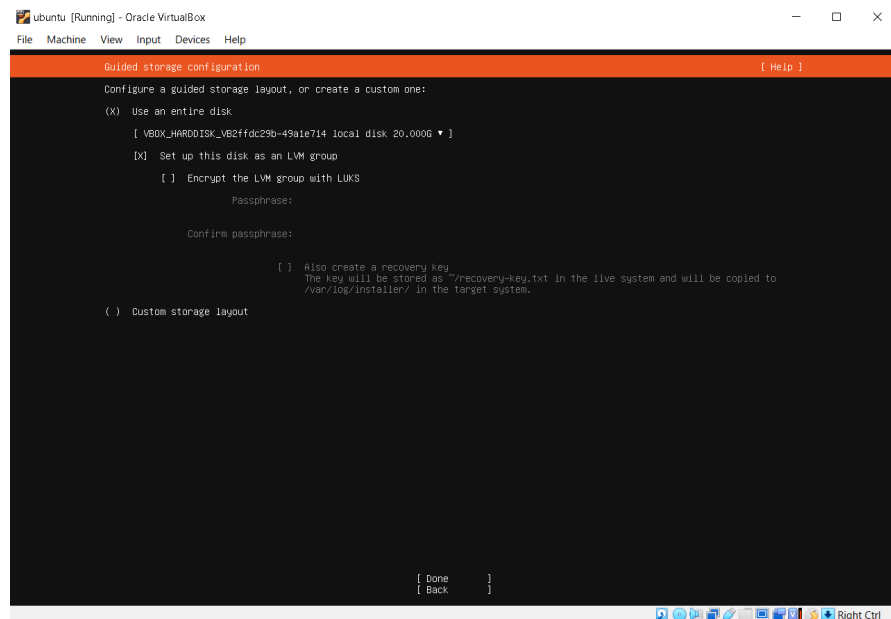


- Screenshot showing the assigned network address as "ubuntu_installer_network.png".

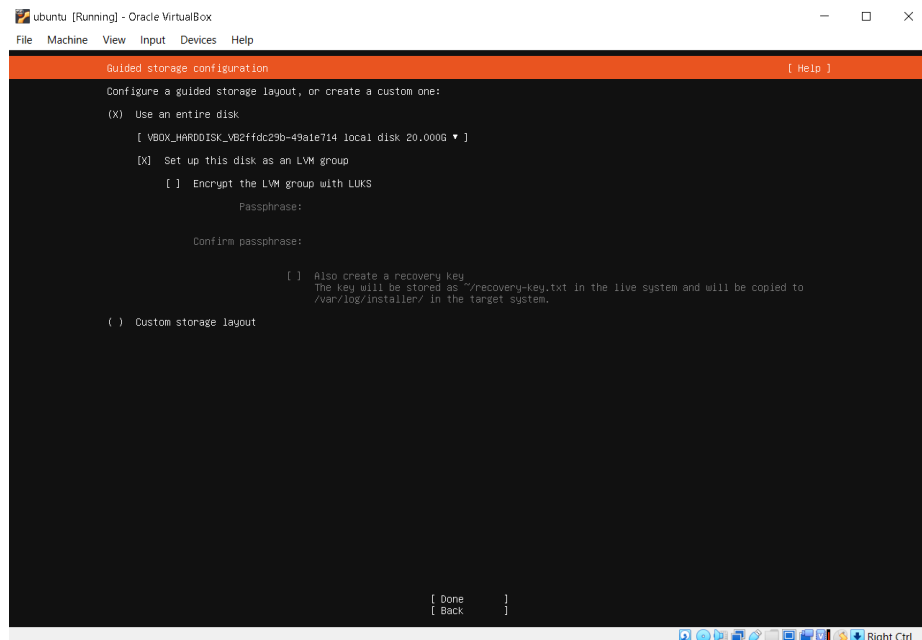
Continue when the network configuration is correct.

Task 9: Configure Storage

- Screenshot showing the selected guided storage option as "ubuntu_guided_storage.png".



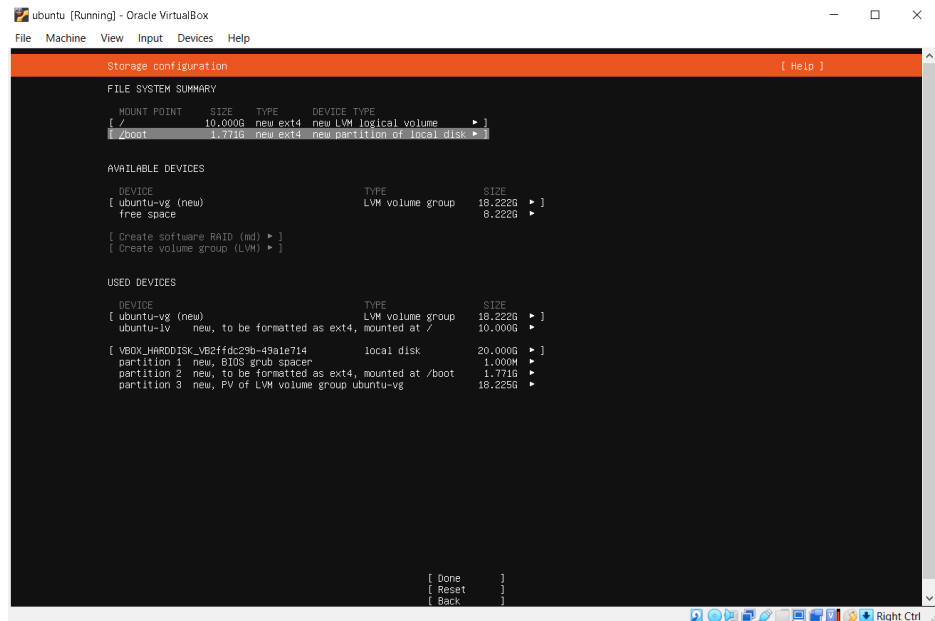
Review the storage configuration before continuing.



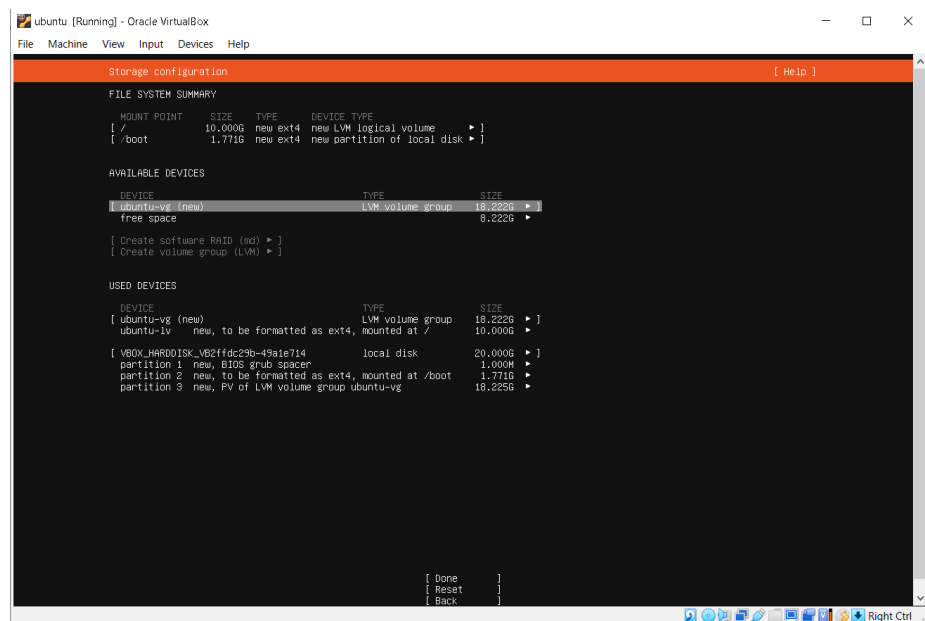
- Screenshot of the storage configuration as "ubuntu_storage_configuration.png".

Task 10: Select the Installation Disk

- Screenshot showing the selected disk as "ubuntu_selected_disk.png".



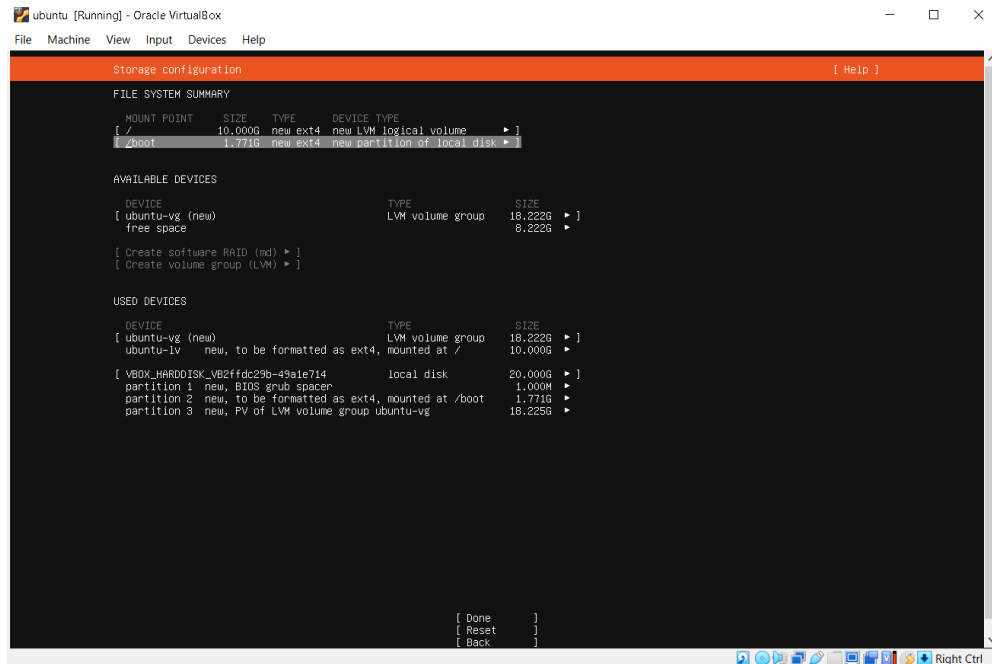
Check the displayed disk size and make sure it is the correct VMware virtual disk.



- Screenshot showing the disk name and size as "ubuntu_disk_details.png".

Task 11: Review the Partitions

- Screenshot showing the proposed partitions as "ubuntu_partition_layout.png".



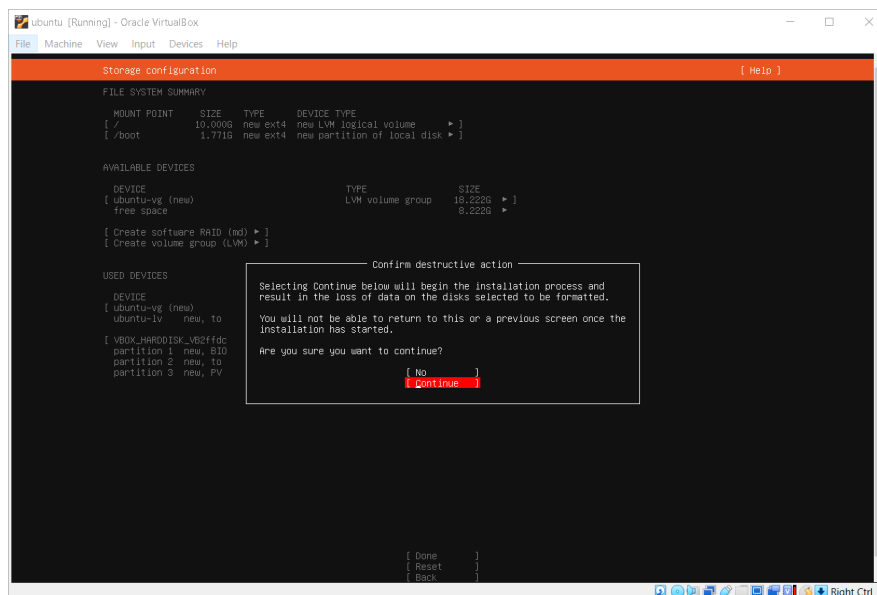
Confirm that the partitions belong to the VMware virtual disk.

Select Done only after reviewing the complete layout.

Task 12: Confirm the Storage Changes

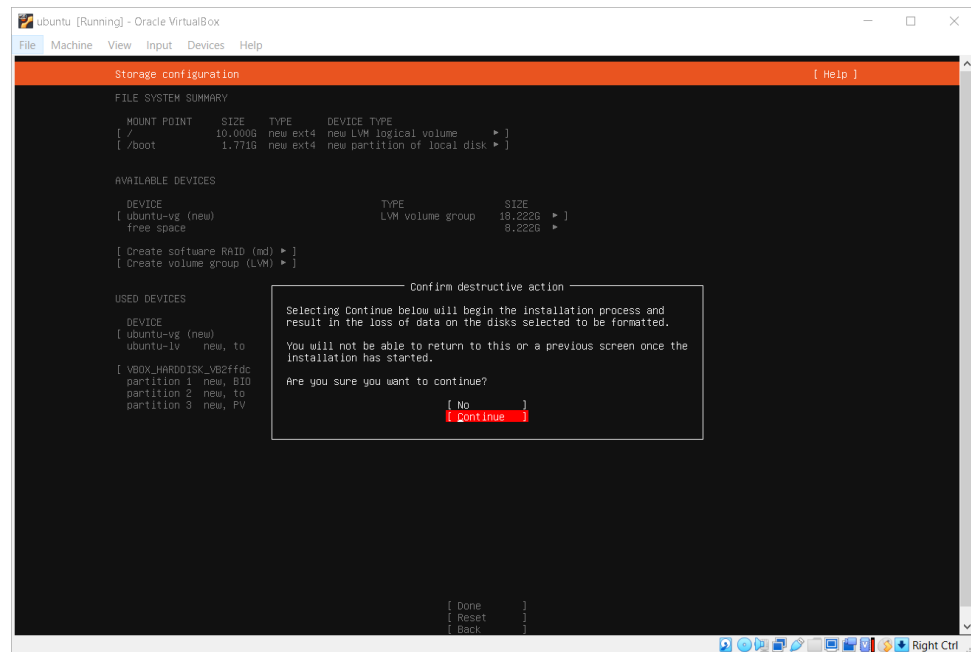
Read the destructive-action warning carefully.

- Screenshot showing the final warning as "ubuntu_storage_warning.png".



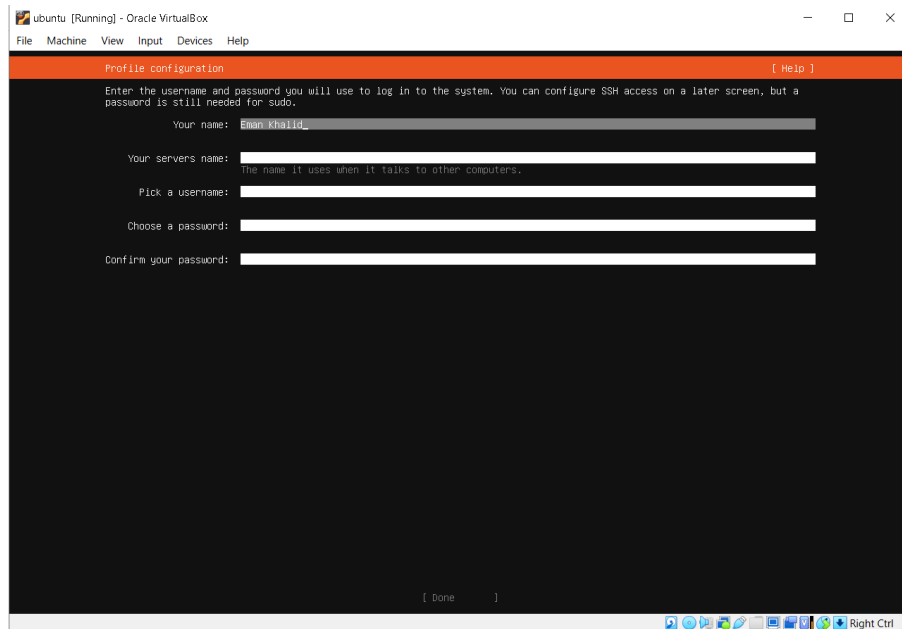
Confirm the changes only when you are certain that the VMware virtual disk is selected.

- Screenshot showing the confirmation selection as "ubuntu_storage_confirmed.png".



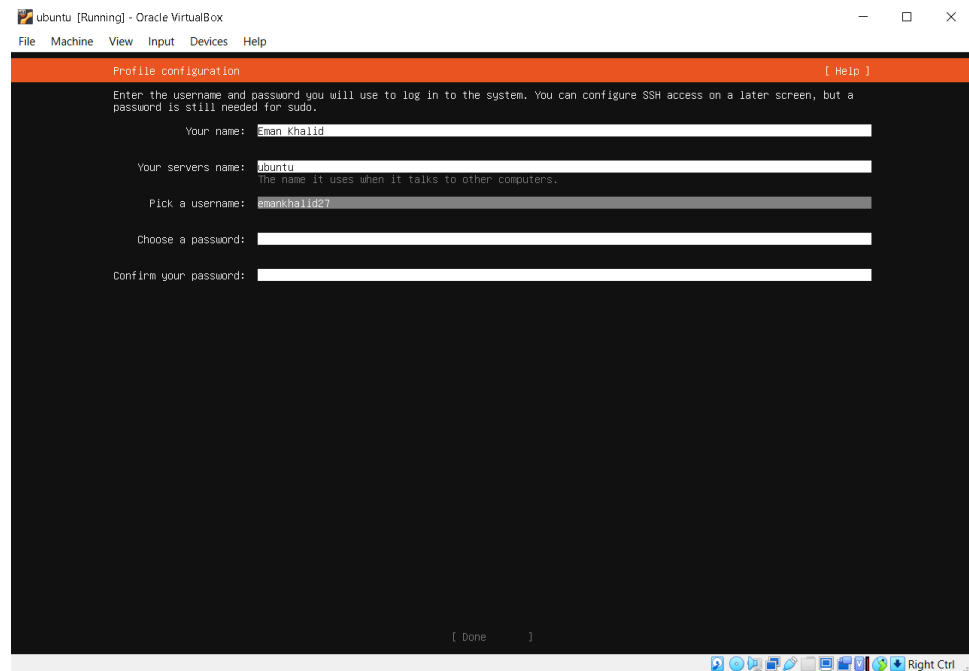
Task 13: Configure the Ubuntu Profile

- Screenshot showing the entered name as "ubuntu_actual_name.png".

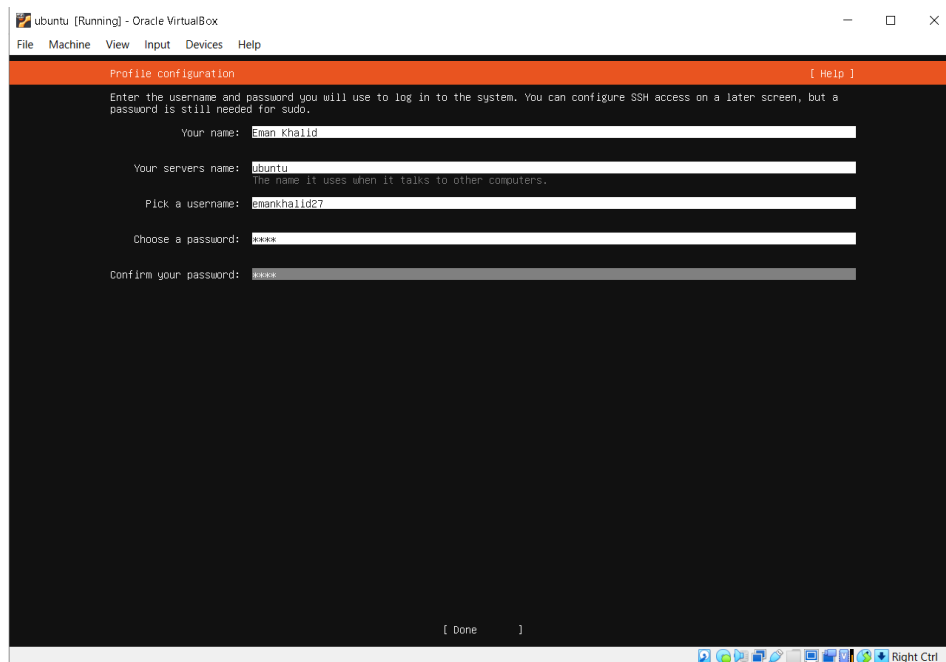


Enter ubuntu in the Your server's name field.

- Screenshot showing the server name as "ubuntu_server_name.png".

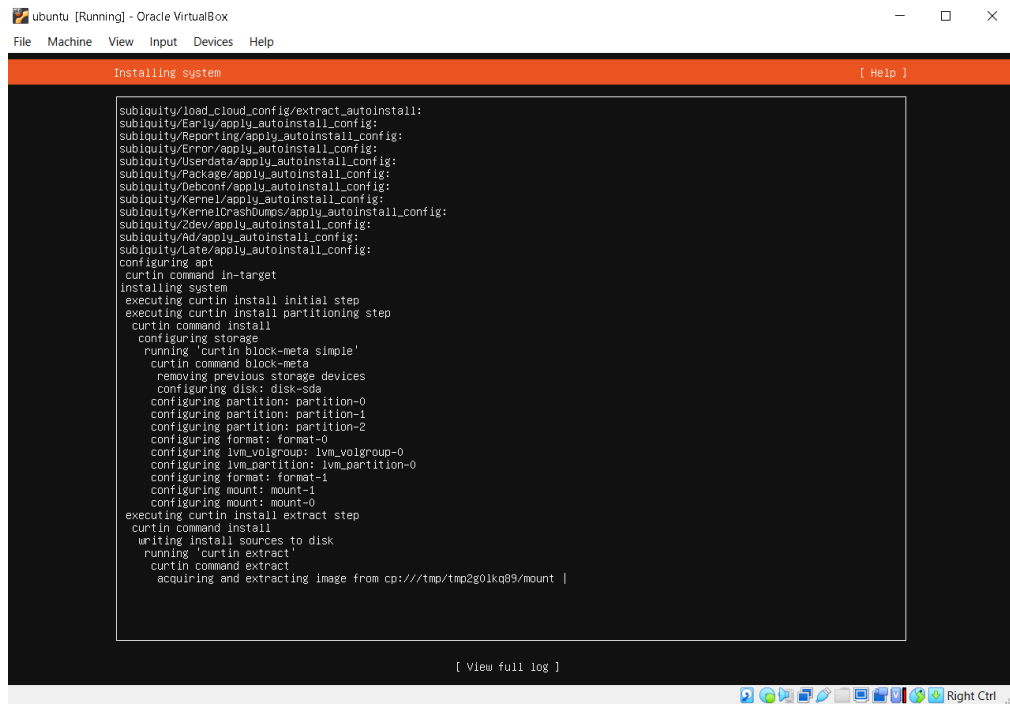


- Screenshot showing the Ubuntu username as "ubuntu_username.png".
- Screenshot of the completed profile page as "ubuntu_profile_setup.png".



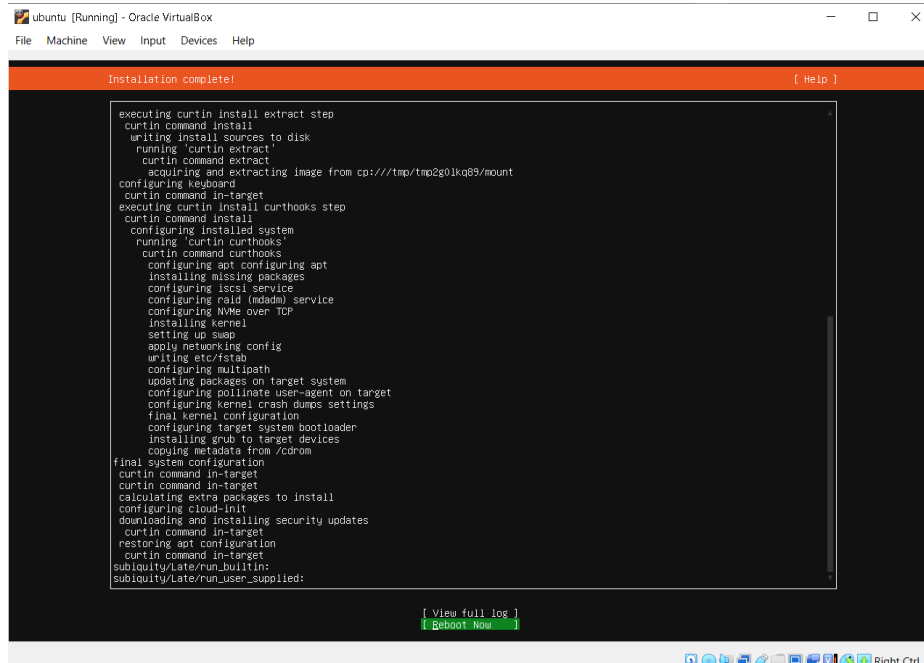
Task 14: Complete the Software Installation

- Screenshot showing the installation progress as "ubuntu_installation_progress.png".



Task 15: Reboot and Verify the Ubuntu Identity

- Screenshot of the installation-complete screen as "ubuntu_installation_complete.png".



- Screenshot showing the Ubuntu login screen as "ubuntu_login_screen.png".

Sign in using the Ubuntu username and password configured in Task 13.

```
ubuntu login: emankhalid27
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)
```

```
* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

System information as of Fri 2 Oct 12:27:52 UTC 2026

```
System load:          0.17
Usage of /:           50.6% of 9.75GB
Memory usage:         13%
Swap usage:           0%
Processes:            128
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe13:f8e1
```

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

emankhalid27@ubuntu:~\$

```
ubuntu login: emankhalid27
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)
```

```
* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

System information as of Fri 2 Oct 12:27:52 UTC 2026

```
System load:          0.17
Usage of /:           50.6% of 9.75GB
Memory usage:         13%
Swap usage:           0%
Processes:            128
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe13:f8e1
```

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

emankhalid27@ubuntu:~\$

- Screenshot showing the successful terminal login as "ubuntu_terminal_login.png".
- Save a clear screenshot showing the complete prompt as "ubuntu_identity_verified.png".

```
emankhalid27@ubuntu:~$
```

Task 16: Find the Ubuntu Server IP Address

- Screenshot showing the command and its output as "ubuntu_ip_addr_command.png".

```
emankhalid27@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:13:f8:e1 brd ff:ff:ff:ff:ff:ff
    altname enx08002713f8e1
    inet 192.168.100.249/24 metric 100 brd 192.168.100.255 scope global dynamic enp0s3
        valid_lft 85919sec preferred_lft 85919sec
    inet6 fe80::a00:27ff:fe13:f8e1/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
emankhalid27@ubuntu:~$
```

- Screenshot clearly showing the correct IPv4 address as "ubuntu_ip_address.png".

```
emankhalid27@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:13:f8:e1 brd ff:ff:ff:ff:ff:ff
    altname enx08002713f8e1
    inet 192.168.100.249/24 metric 100 brd 192.168.100.255 scope global dynamic enp0s3
        valid_lft 85919sec preferred_lft 85919sec
    inet6 fe80::a00:27ff:fe13:f8e1/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
emankhalid27@ubuntu:~$
```

Task 17: Connect to Ubuntu Server from Windows

Keep the Ubuntu Server virtual machine powered on.

Open Command Prompt or PowerShell on Windows.

Run the following SSH command, replacing the placeholders with your Ubuntu username and server IP address:

- Screenshot showing the SSH command as "windows_ssh_command.png".

```
C:\Users\hp>ssh EmanKhalid27@192.168.100.249
The authenticity of host '192.168.100.249 (192.168.100.249)' can't be established.
ED25519 key fingerprint is SHA256:O+663MS42fcQC+f4wklW61Fye9Q9Hf3ersNqA8gVLRw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Type "yes" if you are asked to accept the server fingerprint.

- Screenshot showing the accepted fingerprint as "windows_ssh_fingerprint.png".

```
C:\Users\hp>ssh EmanKhalid27@192.168.100.249
The authenticity of host '192.168.100.249 (192.168.100.249)' can't be established.
ED25519 key fingerprint is SHA256:O+663MS42fcQC+f4wklW61Fye9Q9Hf3ersNqA8gVLRw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

- Screenshot showing the successful SSH session as "windows_ssh_login.png".

```
C:\Users\hp>ssh EmanKhalid27@192.168.100.249
The authenticity of host '192.168.100.249 (192.168.100.249)' can't be established.
ED25519 key fingerprint is SHA256:O+663MS42fcQC+f4wklW61Fye9Q9Hf3ersNqA8gVLRw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.100.249' (ED25519) to the list of known hosts.
EmanKhalid27@192.168.100.249's password:
```

Verify that the remote prompt displays <emankhalid27>@ubuntu.

- Screenshot showing the remote prompt as "windows_ssh_identity.png".

```

C:\Users\hp>ssh emankhalid27@192.168.100.249
emankhalid27@192.168.100.249's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri  2 Oct 13:09:26 UTC 2026

System load:  0.04          Processes:            132
Usage of /:   50.9% of 9.75GB Users logged in:          0
Memory usage: 15%          IPv4 address for enp0s3: 192.168.100.249
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

emankhalid27@ubuntu:~$

```

Task 18: Prepare Lab1 Solution PDF

- Screenshot of the completed title page as "solution_title_page.png".



Fatima Jinnah Women University
Opening Portals of Excellence Through Higher Education

**Lab 01: GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation**

DevOps

Name : Emun Khalid

Reg no : 23-BCS-026

Course Code: CS-649

Section: A

GitHub username: EmunKhalid27

Submitted To: Sir Waqas Saleem

Department: Computer Science